

Otodata Nee-Vo API v33

Modified on October 1, 2026

History

Version	Date	Modified by	Description
1	2016-05-17	Frédéric Courchesne	Initial Document
2	2016-05-19	Frédéric Courchesne	Added “Reliability” section. Added “Redirection” section to the Otodata web portal. “Device” model update.
3	2016-06-14	Frédéric Courchesne	Secondary server URL added Details in each API added
4	2016-06-15	Frédéric Courchesne	“Technology” section modified
5	2017-05-03	Frédéric Courchesne	Properties added to the Device model Otodata Portal URL updated
6	2018-04-18	Frédéric Courchesne	Added a property in Device model
7	2018-08-22	Frédéric Courchesne	Removed firewall restriction Added properties in Device model
8	2018-11-27	Frédéric Courchesne	Added properties in Device model Added model DeviceUpdate
9	2019-01-10	Frédéric Courchesne	Added new requests Added new models
10	2019-02-14	Frédéric Courchesne	Added properties in Device model Added properties in TankLevelLog model
11	2019-05-07	Frédéric Courchesne	Added properties in Device model
12	2020-05-28	Frédéric Courchesne	Added values definitions Added parameters in DownloadReport
13	2020-06-04	Frédéric Courchesne	Added property to Device model Added property to DeviceUpdate model
14	2021-03-15	Frédéric Courchesne	Added HTTP Header section in DeviceUpdate Added method GetTankLevels
15	2022-04-13	Frédéric Courchesne	Added method ResetDevice Added method CreateDispatch Added method CancelDispatch Added properties to Device model Added properties to DeviceUpdate model
16	2022-12-16	Samuel Gagnon	Added property to Device model Added details to UpdateDevice
17	2023-03-13	Patrick Bolduc	Added parameter Sensor in GetDevice, UpdateDevice, CreateDispatch, CancelDispatch and GetDeviceTankLevels
18	2023-04-24	Samuel Gagnon	Added parameter in UpdateDevice Added new models
19	2024-04-29	Frédéric Courchesne	Added property to Device model
20	2024-06-05	Kristopher Gagné	Added multiple methods and models related to return requests and RMAs.
21	2024-07-08	Kristopher Gagné	Changed MaxLengths of models to match the database.
22	2024-12-10	Frédéric Courchesne	Decommissioned the “otodataDemoKey”

23	2025-02-14	Abdul-Rahman Mirza	Added method GetMeters
24	2025-03-07	Kristopher Gagné	Added method UpdateReturnRequest Added new models: ReturnRequestUpdate & UpdateReturnRequestResult
25	2025-06-19	Kristopher Gagné	Added Details in UpdateDevice call Fixed typo in UpdateDevice Error Response
26	2025-06-25	Frédéric Courchesne	Fixed RMA item statuses.
27	2025-07-18	Kristopher Gagné	Added Detail in UpdateDevice call
28	2025-08-18	Kristopher Gagné	Modified UpdateRmaItem call parameters
29	2025-11-17	Kamil Poniedziałek	Modified GetDevice and GetDevices calls with new fields Modified UpdateDevice call with new parameter Added method GetDeviceWaranty
30	2026-03-02	Anthony Rusnak	Modified GetDevice and GetDevices and the Device model to expose new fields
31	2026-04-24	Gerges Tannous	Modified UpdateDevice call with new parameters (LocationName and LocationNumber)
32	2026-09-29	Mohamed Amir Hssini	Added new methods: GetDiagnostic and GetDiagnostics
33	2026-10-01	Kamil Poniedziałek	Added alert profile assignment by name to UpdateDevice and alert profile name to GetDevice for the selected sensor and GetDevices for each returned sensor.

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Introduction

To enable data integration collected by Otodata Remote Tank Level Monitors (RTLM) in a third-party application, a web service (WCF) is available to access the Otodata RTLM readings. The default output format is XML.

To use the JSON output format, the HTTP headers of the request must contain the following option:
`Accept: application/json; charset=utf-8`

To reduce the bandwidth, it is recommended to use dynamic compression by adding the following option in the HTTP headers:
`Accept-Encoding: gzip`

Reliability

To ensure uninterrupted service, the service is hosted on two physical servers belonging to the Otodata Wireless Network. The primary server and the secondary server each have two connectivity links to the Internet (transparent to the user).

Primary Server URLs:	https://neevo.otodata.ca/public/api/v1/DataService.svc https://telematics.otodatanetwork.com:4431/v1.0/DataService.svc
Secondary Server URLs:	https://neevo2.otodata.ca/public/api/v1/DataService.svc https://telematics02.otodatanetwork.com:4431/v1.0/DataService.svc

Security

The security of the web service is divided into two aspects: level of encryption and authentication method.

Encryption

The HTTPS communication protocol is required.

Authentication

Each request must contain a unique API Key in the form of a string of alphanumeric characters. The API Key can be generated or modified on the web portal at the following address:

https://neevo.otodata.ca/#/integrations/apikey

If you are an integrator and do not have access to the web portal, please ask to your client to provide you with the API key.

The API Key can be provided in two different ways:

1. as a parameter in the URL:
   `/v1.0/DataService.svc/devices?k=`***`API_KEY`***

2. added in the HTTP headers:
   `Authorization: Bearer` ***`API_KEY`***

Note that GetDiagnostic and GetDiagnostics methods can only have the API Key supplied through the HTTP headers.

Request definitions

This section contains the queries that can be sent to the web service.

1. GetDevices

Returns a list of modules

URL	`/v1.0/DataService.svc/devices?k={key}&lastDateUtc={lastDateUtc}`
Method	`GET`
URL Parameters	{key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {lastDateUtc} **Optional** ISO8601 date (UTC), example : 2018-11-22T13:00:00Z
Success Reponse	200 – OK Content: array of Device objects (defined on page 30)
Error Responses	400 – Bad Request Parameter {lastDateUtc} does not follow ISO8601 format 403 – Forbidden Parameter {key} is missing or invalid

Example	`GET /v1.0/DataService.svc/devices?k=`***`API_KEY`***
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	
	`GET /v1.0/DataService.svc/devices`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer `***`API_KEY`***

2. GetDevice

Returns a module.

URL	`/v1.0/DataService.svc/devices/{id}?k={key}&sensor={sensor}`
Method	`GET`
URL Parameters	{id} **Required** integer, example: 123456 {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {sensor} **Optional** integer, represents the sensor index of the device. If not provided, default index 0 is used.
Success Reponse	200 – OK Content: Device object (defined on page 30)
Error Responses	400 – Bad Request Parameter {id} is not an integer 403 – Forbidden Parameter {key} is missing or invalid 404 – Bad Request Parameter {id} is missing or invalid

Example	`GET /v1.0/DataService.svc/devices/123456`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

3. GetSampleDevice

Returns a dummy module.

URL	`/v1.0/DataService.svc/devices/sample`
Method	`GET`
Success Reponse	200 – OK Content: Device object (defined on page 30)

Example	`GET /v1.0/DataService.svc/devices/sample`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`

4. GetDeviceWarranties

Returns a module.

URL	`/v1.0/DataService.svc/devices/{id}/warranties?k={key}`
Method	`GET`
URL Parameters	{id} **Required** integer, example: 123456 {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.*
Success Reponse	200 – OK Content: List of DeviceWarranty objects (defined on page 42), ordered from newest to oldest
Error Responses	400 – Bad Request Parameter {id} is not an integer 403 – Forbidden Parameter {key} is missing or invalid 404 – Bad Request Parameter {id} is missing or invalid

Example	`GET /v1.0/DataService.svc/devices/123456/warranties`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

5. UpdateDevice

Modify a module.

URL	`/v1.0/DataService.svc/devices/{id}?k={key}&sensor={sensor}`
Method	`POST`
HTTP Header	Content-Type **Required** `application/json` or `application/xml`
URL Parameters	{id} **Required** integer, example: 123456 {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {sensor} **Optional** integer, represents the sensor index of the device. If not provided, default index 0 is used. {enforceValidation} **Optional** boolean, indicates if the whole update should be canceled if there's at least one given field that can't be updated as requested. Also, if enforceValidation is true and some fields can't be updated, errors details will be returned. If this parameter is not provided, the errors will be silently ignored.
Body Parameters	A DeviceUpdate model. (defined on page 35)
Success Reponse	200 – OK Content: DeviceUpdateResult object (defined on page 35)
Error Responses	400 – Bad Request Parameter {id} is not an integer 403 – Forbidden Parameter {key} is missing or invalid 404 – Not Found Parameter {id} is missing or invalid

Example

```
POST /v1.0/DataService.svc/devices/123456
Host: telematics.otodatanetwork.com:4431
Content-Type: application/json
Accept: application/json; charset=utf-8
Authorization: Bearer API_KEY

{
    "Name": "Demo name for tank #123456",
    "TankFormatName": "Propane 1000 gal"
}
```

Details

a) Each tank format may be associated to a specific product on the Nee-Vo web portal. To be able to assign a tank format to a device, both the tank format and the device must be associated to the same product**,** otherwise changes will be ignored. For example, if the device #12345678 is associated to the product "Propane", then we can set any "Propane" tank format to the device. Also, a tank format not associated to a specific product can be assigned to any device. This is mainly useful to create tank formats for all types of gasoline and diesel.

b) Associating a tank format (**TankFormatName**) to a device automatically sets these properties:
- Capacity
- TankFormType
- MeasureDepth
- MeasureHeight
- MeasureWidth
- Offset
- TareWeight (for scale sensors)
- MaxUllage

c) The properties listed above cannot be updated directly, except for the Capacity. The Capacity can be set if the device is not already associated to a tank format.

d) Updating the device product to a different product unassigns the tank format. All the properties listed above will keep their value even if the tank format as been cleared.

e) Tank formats can be created on the Nee-Vo web portal only.

f) For hydrostatic sensors, the product density must be between 0.3 and 2.0 g/cm³.

g) Associating a tank format (TankFormatName) to a device with hydrostatic pressure sensor type must not have missing dimensions (Depth, Height, Width, Offset, MaxUllage).
h) A tank format (TankFormatName) cannot be associated to a device with sensor type Open-Circuit Cage Belt.
i) When AlertProfileName is provided, the profile must be unique, non-deleted, owned by the API key's company and compatible with the selected sensor. Only that sensor's profile assignment is changed. Omitting the field leaves the assignment unchanged. An invalid name returns 400 Bad Request with an AlertProfileName error and does not save the update, regardless of enforceValidation.

6. ResetDevice

Clear all the customer data from Nee-Vo database for a device.

URL	`/v1.0/DataService.svc/devices/{id}/reset?k={key}`
Method	`POST`
HTTP Header	Content-Type **Required** `application/json` or `application/xml`
URL Parameters	{id} **Required** integer, example: 123456 {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.*
Body Parameters	A JSON object containing a list of property names to exclude. Any property from the Device model can be excluded.
Success Reponse	200 – OK Content: empty
Error Responses	400 – Bad Request Parameter {id} is not an integer 403 – Forbidden Parameter {key} is missing or invalid 404 – Bad Request Parameter {id} is missing or invalid

Example	`POST /v1.0/DataService.svc/devices/123456/reset`
	`Host: telematics.otodatanetwork.com:4431`
	`Content-Type: application/json`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***
	
	`{`
	`  “Exclude”: [“Name”, “TankSerialNumber”]`
	`}`

7. CreateDispatch

Create a dispatch for a device. If an end-user using the Nee-Vo mobile application have enabled notifications, a push notification will be sent. If an integration is enabled, an event will be sent.

URL	`/v1.0/DataService.svc/devices/{id}/createdispatch?k={key}&sensor={sensor}`
Method	`GET`
HTTP Header	Content-Type **Required** `application/json` or `application/xml`
URL Parameters	{id} **Required** integer, example: 123456 {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {sensor} **Optional** integer, represents the sensor index of the device. If not provided, default index 0 is used.
Body Parameters	None.
Success Reponse	200 – OK Content: empty
Error Responses	400 – Bad Request Parameter {id} is not an integer 403 – Forbidden Parameter {key} is missing or invalid 404 – Bad Request Parameter {id} is missing or invalid

Example	`POST /v1.0/DataService.svc/devices/123456/createdispatch`
	`Host: telematics.otodatanetwork.com:4431`
	`Content-Type: application/json`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

8. CancelDispatch

Cancel a dispatch for a device. No push notification is sent to the end-user using the Nee-Vo mobile application when cancelling a dispatch. If an integration is enabled, an event will be sent.

URL	`/v1.0/DataService.svc/devices/{id}/canceldispatch?k={key}&sensor={sensor}`
Method	POST
HTTP Header	Content-Type **Required** `application/json` or `application/xml`
URL Parameters	{id} **Required** integer, example: 123456 {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {sensor} **Optional** integer, represents the sensor index of the device. If not provided, default index 0 is used.
Body Parameters	None.
Success Reponse	200 – OK Content: empty
Error Responses	400 – Bad Request Parameter {id} is not an integer 403 – Forbidden Parameter {key} is missing or invalid 404 – Bad Request Parameter {id} is missing or invalid

Example	`POST /v1.0/DataService.svc/devices/123456/canceldispatch`
	`Host: telematics.otodatanetwork.com:4431`
	`Content-Type: application/json`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***API_KEY***

9. GetTankLevels

Get the historical level readings for all the modules linked to a company. Can be used to get the data for a single module if the optional parameter {id} is provided.

URL	`/v1.0/DataService.svc/tanklevels?k={key}&startDateUtc={startDateUtc}&endDateUtc={endDateUtc}&page={page}`
Method	`GET`
URL Parameters	{key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {startDateUtc} **Required** ISO8601 date (UTC), example: 2018-04-25T08:30:00Z {endDateUtc} **Required** ISO8601 date (UTC), example: 2019-01-10T20:30:00Z {page} **Optional** integer, represents the page index of the results. If not provided, the content of the first page (index 0) is returned. The number of results is limited to 10000 readings per page.
Success Reponse	200 – OK Content: TankLevels object (defined on page 36)
Error Responses	400 – Bad Request Parameter {id} is not an integer 403 – Forbidden Parameter {key} is missing or invalid 404 – Bad Request Parameter {startDateUtc} or {endDateUtc} is missing or invalid

Example	`GET /v1.0/DataService.svc/tanklevels?startDateUtc=2018-05-20T10:00:00Z`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

10. GetDeviceTankLevels

Get the historical level readings for a module.

URL	`/v1.0/DataService.svc/devices/{id}/tanklevels?k={key}&startDateUtc={startDateUtd}&endDateUtc={endDateUtc}&sensor={sensor}`
Method	`GET`
URL Parameters	{id} **Required** integer, example: 123456 {startDateUtc} **Required** ISO8601 date (UTC), example: 2018-04-25T08:30:00Z {endDateUtc} **Required** ISO8601 date (UTC), example: 2019-01-10T20:30:00Z {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {sensor} **Optional** integer, represents the sensor index of the device. If not provided, default index 0 is used.
Success Reponse	200 – OK Content: TankLevels object (defined on page 36)
Error Responses	400 – Bad Request Parameter {id} is not an integer 403 – Forbidden Parameter {key} is missing or invalid 404 – Bad Request Parameter {id}, {startDateUtc} or {endDateUtc} is missing or invalid

Example	`GET /v1.0/DataService.svc/devices/123456/tanklevels?startDateUtc=2018-05-20T10:00:00Z&endDateUtc=2019-01-10T20:30:00Z`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***API_KEY***

11. DownloadReport

Download a summary report in different file formats.

URL	`/v1.0/DataService.svc/downloadReport?k={key}&format={format}&excludeNonInstalled={excludeNonInstalled}&includeSubCompanies={includeSubCompanies}`
Method	`GET`
URL Parameters	{key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {format} **Optional** string, example: CSV {excludeNonInstalled} **Optional** boolean {includeSubCompanies} **Optional** Boolean
Success Reponse	200 – OK Content: Byte stream containing the file 204 – No Content
Error Responses	403 – Forbidden Parameter {key} is missing or invalid

Example	`GET /v1.0/DataService.svc/downloadReport?format=CSV`
	`Host: telematics.otodatanetwork.com:4431`
	`Authorization: Bearer` ***API_KEY***

Note	Supported formats are: 1. Excel (default format) 2. CSV 3. SuburbanSoftware

12. GetReturnRequests

Get Return requests and its associated items (Return request items) linked to a company.

URL	`/v1.0/DataService.svc/returnRequests?k={key}&isOpen={isOpen}&page={pageIndex}`
Method	`GET`
URL Parameters	{key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {isOpen} **Required** Boolean, if true, returns a list of not completed Return Requests. On false, returns a list of completed Return Requests. {page} **Optional** integer, represents the page index of the results. If not provided, the content of the first page (index 0) is returned. The number of results is limited to 100 readings per page in application settings. If the application setting is not found, it will be limited to the default value of 10000 results per pages.
Success Reponse	200 – OK Content: List of ReturnRequest object (defined on page 37)
Error Responses	403 – Forbidden Parameter {key} is missing or invalid

Example	`GET /v1.0/DataService.svc/returnRequests?isOpen=true`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

13. GetReturnRequest

Get a Single Return requests and its associated items (Return request items) linked to a company with provided return request {id} parameter.

URL	`/v1.0/DataService.svc/returnRequests/{id}?k={key}`
Method	`GET`
URL Parameters	{id} **Required** Integer, example: `123456` {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.*
Success Reponse	200 – OK Content: ReturnRequest object (defined on page 37)
Error Responses	400 – Bad Request Parameter {id} is not an integer 403 – Forbidden Parameter {key} is missing or invalid

Example	`GET /v1.0/DataService.svc/returnRequests/123456`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

14. UpdateReturnRequest

Modify the comment of a Return Request.

URL	`/v1.0/DataService.svc/returnRequests/{id}?k={key}`
Method	POST
HTTP Header	Content-Type **Required** `application/json` or `application/xml`
URL Parameters	{id} **Required** alphanumeric, example: RR00001 {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.*
Body Parameters	A ReturnRequestUpdate model. (defined on page 38)
Success Reponse	200 – OK Content: UpdateReturnRequestResult object (defined on page 40)
Error Responses	400 – Bad Request The format of the {id} is invalid 400 – Bad Request Nothing to update 403 – Forbidden Parameter {key} is missing or invalid 403 – Forbidden Client not found

Example	`POST /v1.0/DataService.svc/returnRequests/RR00001`
	`Host: telematics.otodatanetwork.com:4431`
	`Content-Type: application/json`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***
	
	`{`
	`  “Comment”: “Base comment”`
	`}`

15. GetRmas

Get a list of RMAs and its associated items (RMA items) linked to a company.

URL	`/v1.0/DataService.svc/rmas?k={key}&isOpen={isOpen}` `&page={pageIndex}`
Method	`GET`
URL Parameters	{key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {isOpen} **Required** Boolean, if true, returns a list of not completed Return Requests. If false, returns a list of completed Return Requests. {page} **Optional** integer, represents the page index of the results. If not provided, the content of the first page (index 0) is returned. The number of results is limited to 100 readings per page in application settings. If the application setting is not found, it will be limited to the default value of 10000 results per pages.
Success Reponse	200 – OK Content: List of Rma object (defined on page 39)
Error Responses	403 – Forbidden Parameter {key} is missing or invalid

Example	`GET /v1.0/DataService.svc/rmas?`
	`isOpen=true`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

16. GetRma

Get a Single RMA and its associated items (RMA items) linked to a company with provided RMA {id} parameter.

URL	`/v1.0/DataService.svc/rmas/{id}?k={key}`
Method	`GET`
URL Parameters	{id} **Required** string, example: RMA123 {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.*
Success Reponse	200 – OK Content: Rma object (defined on page 39)
Error Responses	403 – Forbidden Parameter {key} is missing or invalid 403 – Forbidden Parameter {id} is missing or invalid

Example	`GET /v1.0/DataService.svc/rmas/RMA123`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

17. UpdateRmaItem

Modify an RMA item.

URL	`/v1.0/DataService.svc/rmaItems/{rmaId}/devices/{deviceId}?k={key}`
Method	POST
HTTP Header	Content-Type **Required** `application/json` or `application/xml`
URL Parameters	{rmaId} **Required** string, example: RMA001 {deviceId} **Required** Integer, example: 12345678 {key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.*
Body Parameters	A RmaItemUpdate model. (defined on page 40)
Success Reponse	200 – OK Content: UpdateRmaItemResult object (defined on page 40)
Error Responses	400 – Bad Request The format of the {id} is invalid 400 – Bad Request Nothing to update 400 – Bad Request Rma id is missing 403 – Forbidden Parameter {key} is missing or invalid

Example	`POST /v1.0/DataService.svc/rmaItems/RMA001/devices/ 123456`
	`Host: telematics.otodatanetwork.com:4431`
	`Content-Type: application/json`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***
	`{`
	`“IsReconciled”: true,`
	`“CustomerComment”: “Base comment”`
	`}`

18. GetGasMeters

Gets the historical level readings for all the modules linked to a company that have a gas meter sensor. Gets all devices with gas meters.

URL	`/v1.0/DataService.svc/meters?k={key}&startDateUtc={startDateUtc}&endDateUtc={endDateUtc}&page={pageIndex}`
Method	GET
URL Parameters	{key} **Required** alphanumeric *Can be set in the HTTP headers instead, see the* Authentication *section.* {startDateUtc} **Required** ISO8601 date (UTC), example: 2025-01-15. {endDateUtc} **Required** ISO8601 date (UTC), example: 2025-02-15. {page} **Optional** integer, represents the page index of the results. If not provided, the content of the first page (index 0) is returned. The number of results is limited to 10000 meters per page.
Success Reponse	200 – OK Content: GasMeters object (defined on page 40)
Error Responses	403 – Forbidden Parameter {key} is missing or invalid 404 – Bad Request Parameter {startDateUtc} or {endDateUtc} is missing or invalid

Example	`GET /v1.0/DataService.svc/meters?startDateUtc=2018-05-20T10:00:00Z`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

19. GetDiagnostic

Returns the canonical diagnostic result for a module linked to the company authorized by the API key. Note that the API Key can only be added in the HTTP Header for this method.

URL	`/v1.0/DataService.svc/diagnostics/{serialNumber}`
Method	`GET`
HTTP Header	Authorization **Required** `Bearer API_KEY` Accept-Language **Optional** Show results in the selected language . Defaults to English. *Note that only English and French are currently supported*
URL Parameters	{serialNumber} **Required** Int64, example: 200000027
Success Response	200 - OK Content: DiagnosticResult object (defined in the Model definitions section)
Error Responses	400 - Bad Request Parameter {serialNumber} is not an integer 403 - Forbidden API Key is missing or invalid 404 - Not Found Device is outside the permitted company scope or has no diagnostic result

Example	`GET /v1.0/DataService.svc/diagnostics/200000027`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

20. GetDiagnostics

Returns diagnostic results for modules in the company scope authorized by the API key. Results can be filtered by trouble type and route name.
Note that the API Key can only be added in the HTTP Header for this method.

URL	`/v1.0/DataService.svc/diagnostics?page={page}&troubleType={troubleType}&routeName={routeName}`
Method	`GET`
HTTP Header	Authorization **Required** `Bearer API_KEY` Accept-Language **Optional** Show results in the selected language. Defaults to English. *Note that only English and French are currently supported*
URL Parameters	{page} **Optional** integer, one-based page index. If omitted or lower than 0, the first page is returned. Results are limited to 100 diagnostics per page. {troubleType} **Optional** string. Supported names are Hardware, Communication, and Sensor (case-insensitive). If omitted, diagnostics with any of these trouble categories are returned. {routeName} **Optional** string. Filters results to devices associated with the specified route name.
Success Response	200 - OK Content: DiagnosticsDetailsDto object (defined in the Model definitions section)
Error Responses	400 - Bad Request Parameter {troubleType} is invalid 403 - Forbidden API Key is missing or invalid 500 - Internal Server Error An unexpected server error occurred

Example	`GET`
	`/v1.0/DataService.svc/diagnostics?page=0&troubleType=Hardware&routeName=Main`
	`Host: telematics.otodatanetwork.com:4431`
	`Accept: application/json; charset=utf-8`
	`Authorization: Bearer` ***`API_KEY`***

Model definitions

This section contains the models that can be received from the web service.

1. Device

Reference for CorrectedVolume property:
https://www.ic.gc.ca/eic/site/mc-mc.nsf/eng/lm00136.html

Property	Type	Maximum Length	Description	Unit
Address	String	256	Civic number and street name	
BatteryAlarm	Boolean		**false** = battery OK **true** = low battery	
BatteryLevel	Double		Battery voltage	Volts
Capacity	Int32		Capacity of the tank (s)	Liters
City	String	128	City	
CorrectedVolume	Int32		Represents the volume of the tank corrected for temperature. The correction data was taken from Measurement Canada's website (URL above this table). **This only applies to propane.**	Liters
Country	String	128	Country	
CriticalLowAlarmThreshold	Double		Threshold determining the critical low level of a tank	
CustomerId	String	128	Customer ID, labelled as Account Number on the web portal	
DatabaseId	String	64	Custom field where customer can input additional data	

Property	Type	Maximum Length	Description	Unit
Email	String	256	Customer's email address	
EmptyAlarmThreshold	Double		Threshold determining the empty level of a tank	
AlertProfileName	String	128	Name of the alert profile assigned to the sensor represented by this Device object; null if none.	
GeneralLedgerLocation	String	64	Custom field where customer can input additional data	
GpsLatitude	Double		GPS coordinate reported by the device	
GpsLongitude	Double		GPS coordinate reported by the device	
HighDrainAlarm	Boolean		**false** = tank OK **true** = the tank empties quickly	
HoursToEmpty	Int32		Estimated time before run-out	Hours
HoursToOptimalFill	Int32		Estimated time before reaching the optimal filling level	Hours
Id	Int64		Otodata module serial number	
Inventory	Double		Volume of product in the tank	Liters
IsInstalled	Boolean		Indicates if the module is installed	
InstalledDate	DateTime		The date the device was installed	
Last24hr	Double		Last 24 hours consumption	Liters
Last48hr	Double		Last 48 hours consumption	Liters
Last72hr	Double		Last 72 hours consumption	Liters
LastFill	DateTime		Date & time of the last fill **Note: A date equals to 1900-01-01 indicates that no statement is available**	UTC
LastLevel	Double		Tank level from 0.0000 to 1.0000	%

Property	Type	Maximum Length	Description	Unit
LastWeek	Double		Last week of consumption	
LastRead	DateTime		Date & time of the last communication **Note: A date equals to 1900-01-01 indicates that no statement is available**	UTC
Latitude	Double		GPS coordinate of the address	
Height	Double		Height of the product in the tank	Centimeters
ProductTemperature	Double		Temperature of the product inside the tank	Celsius
Longitude	Double		GPS coordinate of the address	
LowAlarmThreshold	Double		Threshold determining the low level of a tank	
MeasureDepth	Double		Tank's depth	Centimeters
MeasureHeight	Double		Tank's height	Centimeters
MeasureWidth	Double		Tank's width	Centimeters
Model	String	128	Model name of the RTLM	
Name	String	128	Client name	
Note	String	2048	Additional customer data	
Offset			Distance between the base of the ultrasonic sensor and the top of the tank. **Note: Applies to ultrasonic and hydrostatic pressure sensors.**	Centimeters
OverfillAlarmThreshold			Threshold determining the overfill level of a tank	
PostalCode	String	32	Postal code	
Pressure	Double		Pressure measured for the tank	Kilopascals
Product	String	128	Stored product in tank. See section Product on page 44 for more information.	
ReadingTrouble	Boolean		**false** = the statement is valid	

Property	Type	Maximum Length	Description	Unit
			true = the statement should not be considered valid	
Region	String	128	Province / State	
Route	String	128	Name of the route to which the tank is associated	
SensorIndex	Byte		The zero-based index of the sensor reporting the reading, if more than one sensor is attached to the device.	
SensorTrouble	Int32 (SensorTrouble)		Trouble with the sensor. See section SensorTrouble on page 43 for more information.	
SignalStrength	Int32		Quality of received radio signal, from 0 to 100	%
Status	String	32	Actual status of the tank. See section Status on page 43 for more information.	
Supplier	String	7	Character string attached to "OTODATA"	
TankFormType	Int32		Tank's shape. See section TankFormType on page 44 for more information.	
TankFormatName	String	128	Name of the tank format to which the tank is associated	
TankName	String	128	Name of the tank	
TankNumber	String	64	Number of the tank	
TankSerialNumber	String	64	Tank serial number	
Temperature	Double		Internal temperature	Celsius
Type	String	1	V = vertical H = horizontal U = ultrasonic	
Ullage	Int32		Filling potential	Liters
Weight	Double		Weight of the tank's content	Kilograms
IsOnCall	Boolean		Indicates if the tank is configured as *Fill on Call*	

Property	Type	Maximum Length	Description	Unit
			(customer calls in for deliveries) **- false** – normal delivery **- true** – Fill on Call	
Value	Double		Raw value reported by the device	
ValueType	Byte		Raw value type. See section ValueType on page 45 for more information.	
LocationName	String	128	Name of the location associated to the tank	
LocationNumber	String	32	Number of the location associated to the tank	
EstimatedDailyDrawRate	Double		An estimation of the daily draw rate based on last week values	%

2. DeviceUpdate

Property	Type	Maximum Length	Description	Unit
LocationNumber	String	32	Location NUmber	
LocationName	String	128	Location Name	
Address	String	256	Civic number and street name	
Capacity	Int32		Capacity of the tank	Liters
City	String	128	City	
Country	String	128	Country	
CustomerId	String	128	Customer ID, labelled as Account Number on the web portal	
DatabaseId	String	64	Custom field where customer can input additional data	
Email	String	256	Customer's email address	
GeneralLedgerLocation	String	64	Custom field where customer can input additional data	
IsInstalled	Boolean		Indicates if the module is installed	
Latitude	Double		GPS coordinate	
Longitude	Double		GPS coordinate	

Property	Type	Maximum Length	Description	Unit
MeasureDepth	Double		Tank's depth	Centimeters
MeasureHeight	Double		Tank's height	Centimeters
MeasureWidth	Double		Tank's width	Centimeters
Name	String	128	Client name	
Note	String	2048	Additional customer data	
Offset			Distance between the base of the ultrasonic sensor and the top of the tank. **Note: Applies to ultrasonic and hydrostatic pressure sensors.**	Centimeters
PostalCode	String	32	Postal code	
Product	String		Stored product in tank. See section Product on page 44 for more information.	
Region	String	128	Province / State	
Route	String	128	Name of the route to which the tank is associated	
TankFormType	Int32		Tank's shape. See section TankFormType on page 44 for more information.	
TankFormatName	String	128	Name of the tank format to which the tank is associated	
TankName	String	128	Name of the tank	
TankNumber	String	64	Number of the tank	
TankSerialNumber	String	64	Tank serial number	
AlertProfileName	String	128	Optional. Unique, non-deleted alert profile name belonging to the API key's company and compatible with the selected sensor. Omit to keep the current assignment; an empty name is invalid.	
IsOnCall	Boolean		Sets whether the tank is in "Fill on Call" mode (true) or not (false)	

3. DeviceUpdateResult

Property	Type	Maximum Length	Description	Unit
Errors	Array		Array of UpdateError	

4. UpdateError

Property	Type	Maximum Length	Description	Unit
Property	String		If any, the name of the property that triggered the error	
Message	String		Error description	

5. DeviceWarranty

Property	Type	Maximum Length	Description	Unit
WarrantyType	String		Type of warranty (see page 47 for more details)	
WarrantyStatus	String		Current state of the warranty (see page 46 for more details)	
WarrantyStartDate	DateTime		Warranty start date	UTC
WarrantyEndDate	DateTime		Warranty end date	UTC

6. TankLevels

Property	Type	Maximum Length	Description	Unit
Count	Int32		Number of items in the Logs array	
EndDateUtc	DateTime		Ending date and time of the period.	UTC
Id	Int64		Otodata module serial number	
Logs	Array		Array of TankLevelLog	
StartDateUtc	DateTime		Starting date and time of the period.	UTC
Supplier	String	7	Character string attached to "OTODATA"	

7. TankLevelLog

Property	Type	Maximum Length	Description	Unit
BatteryLevel	Float		Battery voltage	Volts
Id	Int64		Otodata module serial number	
Level	Float		Tank level from 0.0000 to 1.0000	%
LogDateUtc	DateTime		Date of the level reading	UTC
SensorTrouble	Byte		Indicates if there is a sensor trouble.	
Temperature	Float			Celsius
Value	Double		Raw value reported by the device	
ValueType	Byte		Raw value type. See section ValueType on page 45 for more information.	
SignalStrength	Int16		Quality of received radio signal, from 0 to 100	%

8. ReturnRequest

Property	Type	Maximum Length	Description	Unit
CompanyName	String	128	Name of the company.	
ReturnRequestNumber	String	32	The return request's number.	
RmaNumber	String	32	The RMA number.	
CreatedDate	DateTime		Date of creation.	UTC
DeviceCount	Int32		Number of devices in the request (return request items).	
Status	Byte		The actual Return Request Status. See section ReturnRequestStatus on page 46 for more information.	
RequesterName	String	128	The name of the requester.	
RequesterEmail	String	256	The email of the requester.	
Comment	String	MAX	Comment on the return request.	

Property	Type	Maximum Length	Description	Unit
StreetAddress	String	256	Street Adress of the return.	
City	String	128	City of the return.	
PostalCode	String	32	Postal Code of the return.	
Region	String	128	Region of the return.	
Country	String	128	Country of the return.	
ReturnRequestItems	List		List of ReturnRequestItems.	

9. ReturnRequestItem

Property	Type	Maximum Length	Description	Unit
SerialNumber	Int64		Device Serial Number	
Status	Byte		The actual Return Request Item Status. See section ReturnRequestItemStatus on page 46 for more information.	
SensorStatus	String	32	The actual Propane Report Status. See section Status on page 43 for more information.	
SensorTrouble	Byte		Indicates if there is a sensor trouble. See page 43 for more information.	
LastActivityDate	DateTime		The Device's last activity date (sensor).	UTC
WarrantyStatus	Byte		The actual device's battery warranty status. See section WarrantyStatus on page 46 for more information.	
RefusalReason	String	2048	The refusal reason (if return refusal happens).	

10. ReturnRequestUpdate

Property	Type	Maximum Length	Description	Unit
Comment	String	2048	The Comment on the return request.	

11. UpdateReturnRequestResult

Property	Type	Maximum Length	Description	Unit
Errors	Array		Array of UpdateError	

12. Rma

Property	Type	Maximum Length	Description	Unit
RmaNumber	String	32	The RMA Number.	
CreatedDate	DateTime		The RMA Creation date.	UTC
IsComplete	bool		The status of the RMA.	
RmaItems	List		List of RmaItem.	

10. RmaItem

Property	Type	Maximum Length	Description	Unit
UpdatedDate	DateTime		The RMA item updated date.	UTC
SerialNumber	Int64		The device serial number.	
WarrantyStatus	Byte		The actual device's battery warranty status. See section WarrantyStatus on page 46 for more information.	
Status	Byte		The actual RMA item Status. See section RmaItemStatus on page 46 for more information.	
Conclusion	String	32	The disposition taken with the RMA item.	
Trouble	String	32	The Trouble with the RMA item (if there was any).	
Replacement	Int64		The replacement device's serial number.	
ReturnAddress	String	256	The address of the return.	

Property	Type	Maximum Length	Description	Unit
IsReconciled	Boolean		The status if the RMA item is reconciled.	
CustomerComment	String	2048	Customer comment about the RMA item.	
Route	String	128	The Device's Route name	

11. RmaItemUpdate

Property	Type	Maximum Length	Description	Unit
IsReconciled	Boolean		The status if the RMA item is reconciled.	
CustomerComment	String	2048	Customer comment about the RMA item.	

12. RmaItemUpdateResult

Property	Type	Maximum Length	Description	Unit
Errors	Array		Array of UpdateError	

13. GasMeters

Property	Type	Maximum Length	Description	Unit
Count	Int32		Number of items in the Logs array	
EndDateUtc	DateTime		Ending date and time of the period.	UTC
Meters	Array		Array of GasMeterLog	
StartDateUtc	DateTime		Starting date and time of the period.	UTC
Supplier	String	7	Character string attached to "OTODATA"	

14. GasMeterLog

Property	Type	Maximum Length	Description	Unit
AccountNumber	String		Customer account number	
TankNumber	String		Tank number	
IndexInitialValue	Double		The meter index at the beginning of the billing period	
IndexFinalValue	Double		The meter index at the end of the billing period	
PulseRate	Double		The pulse rate of the meter	
SerialNumber	Int64		Serial number of the gas meter	
Units	String		Unit of measurement	Cubic feet, cubic meters
Volume	Double		Volume of gas consumed during the billing period	*Specified in "Units"*

15. DiagnosticDetails

Property	Type	Maximum Length	Description	Unit
SerialNumber	Int64		Otodata module serial number	
WarrantyStatus	Byte?		Numeric code indicating the device warranty status	
TroubleType	String		Indicates whether the diagnostic result contains a communication sensor or hardware trouble	
Diagnostics	String		Description of the detected diagnostic condition	
CorrectiveAction	String		Recommended steps to resolve the detected issue	
HasRequestOfService	String		Indicates whether a service request currently exists	
Route	String		Name of the route assigned to the device	
AccountNumber	String		Customer account number associated with the device	

Property	Type	Maximum Length	Description	Unit
Address	String		Customer address	
City	String		Customer city	
ProvinceState	String		Province or state	
LastCommunication	String		Last device communication time	
ServiceCallNumber	String		Related service call number	

Value definitions

This section contains a definition of the values that can be assigned to different properties.

1. SensorTrouble

This property is an integer representing sensor related problem. Valid values are:

0 = None
1 = Sensor overload
2 = Value under specs
3 = Value over specs
4 = Sensor unreadable
5 = Sensor accuracy
6 = Invalid parameter configuration
7 = Incompatible sensor connected
8 = Sensor supervision lost

Note 1: A level of 0% can be considered as *Value under specs*.
Note 2: A level of 100% can be considered as *Value over specs*.

2. Status

This property is a text value representing the status of a tank. Valid values are:

- OK
- LOW ALARM
- CRITICAL LOW ALARM
- OVERFILL ALARM
- EMPTY ALARM
- RAPID DRAW
- FILL DETECTION
- COMM TROUBLE
- UNINSTALLED
- INACTIVITY DETECTION
- N/A

3. TankFormType

This property is an integer representing the tank format. Valid values are:

0 = Horizontal Cylinder
1 = Vertical Cylinder
2 = Rectangle
3 = *Reserved – Do not use*
4 = Vertical Oval

4. Product

This property is a text value representing the product stored in the tank. Supported values are:

- Propane
- LPG
- Diesel
- Clear Diesel
- Clear Diesel Efficient
- Clear Diesel Premium
- Dyed Diesel
- Dyed Diesel Efficient
- Dyed Diesel Premium
- Gasoline
- Regular Gasoline
- Super Gasoline
- Jet Fuel
- Kerosene
- Def
- Heating Oil
- Lubricant
- Butane
- Water
- Fertilizer

This property can be set to a custom value using the UpdateDevice API.

5. ValueType

The property "ValueType" represent the kind of value stored in our database. It is needed to understand the meaning of the property "Value".

We always return the value as it is stored in our databases and a conversion may be required to obtain a meaningful output.

Value Type	Definition	Boundaries		Interpretation
		minimum	*maximum*	
0	Percentage	0	10000	The value represents one hundredth of a percent. *Conversion needed to get percentage:* $value \div 100 = x\ \%$
1	Distance	0	16777215	The value represents a distance in millimeters, from 1 millimeter to 16 kilometers. *To convert into meters:* $value \div 1000 = x\ meters$ *To convert into feets:* $value \div 305 = x\ feet$
2	Quantity	0	16777215	The value represents a counter.
3	Temperature	0	65535	The value represents a temperature in deciKelvin, from 0 °K to 6553.5 °K. *To get °Celsius:* $\frac{value}{10} - 273.15 = x\ °C$ *Conversion needed to get °Fahrenheit:* $\left(\frac{value}{10} - 273.15\right) \times \frac{9}{5} + 32 = x\ °F$

6. ReturnRequestStatus

This property represents the status of a return request.

0 = Pending
1 = Analyzing
2 = Completed
3 = RmaReady

7. ReturnRequestItemStatus

This property represents the status of an item in a return request.

0 = Pending
1 = Accepted
2 = Refused

8. RmaItemStatus

This property represents the status on an item in an RMA.

0 = Pending Analysis
1 = Being Analyzed
2 = Complete
3 = Pending Receipt
4 = Not Received
5 = Ignored

9. WarrantyStatus

This property represents a Device's battery Warranty Status. The value is displayed are based on the WarrantyStartDate and the WarrantyEndDate.

0 = Unknown – *Obsolete*
1 = NoWarranty – *Obsolete*
2 = NotStarted
3 = Expired
4 = Active
5 = BatteryWarrantyOnly – *Obsolete*
6 = Voided
7 = Archived

10. WarrantyType

This property represents the type of warranty associated with a device.

0 = Standard
1 = Extended
2 = BatteryPackOnly

Redirection to the Otodata Nee-Vo web portal

You can access the configuration page of any module using the following URL format:

https://neevo.otodata.ca/#/devices/{id}

The {id} parameter represents the serial number of the module.